Grade 7 - Module 5 Comprehensive Test - TEACHER ANSWER KEY

Scheduled: 2027-03-26 | Covers all four module weeks and all five subjects

Math: 7.SP.1-4 | ELA: W.7.2,7-9; RI.7.1-9 | Science: E.7.9A.1-3 | Social Studies: MS CCRS SS Grade 7 - Byzantine/Islamic civilizations | Spanish: MS World Languages Framework 2016 - Novice communication

Award points by item. Accept equivalent correct mathematical forms and supported written interpretations. For open responses, use the stated rubric; do not require identical wording. Score each subject separately, then apply your grading policy.

Math - 28 points

Score: _____ / 28

1. A survey of 50 students is used to estimate preferences of 600 students. Identify the population and sample. (1 point)

Answer / scoring: Population: all 600 students; sample: the 50 surveyed.

2. A survey asks only soccer players about the most popular school sport. Explain a likely bias. (1 point)

Answer / scoring: Soccer players are overrepresented; they may prefer soccer more than the whole school.

3. Choose a better random sample: first 20 arrivals or 20 names drawn from all student names. (1 point)

Answer / scoring: 20 names drawn from all student names.

4. In a random sample, 15 of 60 students choose art. Estimate how many of 400 students choose art. (1 point)

Answer / scoring: 100

5. Does a random sample guarantee a perfectly correct estimate? (1 point)

Answer / scoring: No; random sampling reduces selection bias but sampling variation remains.

6. Design a representative sample to ask 300 students about lunch choices. (2 points)

Answer / scoring: Use a random selection from all students or random samples across grades; avoid a convenience group. One point for random selection, one for coverage.

7. A survey of 50 students is used to estimate preferences of 600 students. Identify the population and sample. (1 point)

Answer / scoring: Population: all 600 students; sample: the 50 surveyed.

8. A survey asks only soccer players about the most popular school sport. Explain a likely bias. (1 point)

Answer / scoring: Soccer players are overrepresented; they may prefer soccer more than the whole school.

9. Choose a better random sample: first 20 arrivals or 20 names drawn from all student names. (1 point)

Answer / scoring: 20 names drawn from all student names.

10. In a random sample, 15 of 60 students choose art. Estimate how many of 400 students choose art. (1 point)

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11. Does a random sample guarantee a perfectly correct estimate? (1 point)

Answer / scoring: No; random sampling reduces selection bias but sampling variation remains.

12. Design a representative sample to ask 300 students about lunch choices. (2 points)

Answer / scoring: Use a random selection from all students or random samples across grades; avoid a convenience group. One point for random selection, one for coverage.

13. Data A: [6, 8, 8, 10, 13]. Find the mean. (1 point)

Answer / scoring: 9.0

14. For Data A above, find the median. (1 point)

Answer / scoring: 8

15. For Data A above, find the range. (1 point)

Answer / scoring: 7

16. Data B: [7, 7, 8, 11, 12]. Find the median. (1 point)

Answer / scoring: 8

17. An outlier of 100 is added to 2, 3, 4, 5. Which changes more: mean or median? (1 point)

Answer / scoring: Mean. Mean goes from 3.5 to 22.8; median goes from 3.5 to 4.

18. Compare 2, 4, 6 and 4, 4, 4. Find both means and explain their variability. (2 points)

Answer / scoring: Both means are 4. First range is 4; second range is 0. Equal center can hide different spread.

19. Data A: [2, 4, 4, 6, 9]. Find the mean. (1 point)

Answer / scoring: 5.0

20. For Data A above, find the median. (1 point)

Answer / scoring: 4

21. For Data A above, find the range. (1 point)

Answer / scoring: 7

22. Data B: [3, 3, 4, 7, 8]. Find the median. (1 point)

Answer / scoring: 4

23. An outlier of 100 is added to 2, 3, 4, 5. Which changes more: mean or median? (1 point)

Answer / scoring: Mean. Mean goes from 3.5 to 22.8; median goes from 3.5 to 4.

24. Compare 2, 4, 6 and 4, 4, 4. Find both means and explain their variability. (2 points)

Answer / scoring: Both means are 4. First range is 4; second range is 0. Equal center can hide different spread.

ELA - 16 points

Score: _____ / 16

Two Sources for a Forecast

Fictional classroom sources: Source A, "Weather Log," by a student team, dated March 8, records local temperature, wind direction, and pressure at noon for five days. Source B, an unsigned advertisement dated March 8, says that one device "always predicts weather perfectly" but gives no measurements. A research group prefers Source A for describing what was actually observed. It still checks the team's instruments and method, because named authors and a date do not guarantee accuracy. The group writes a cautious forecast, separating data from inference. Its notes include each source's title, creator, date, and the particular information used. In its final explanation, it paraphrases the data instead of copying a paragraph and identifies its source. Readers can judge the explanation more easily because the evidence and its limits are visible.

1. Write a focused research question connected to the passage. (1 point)

Answer / scoring: Accept a specific, answerable question connected to the text, not an unlimited topic.

2. Name two different kinds of sources that could help answer it. (1 point)

Answer / scoring: Accept relevant books/articles, maps/data, interviews/primary accounts; each must suit the question.

3. A source makes a strong claim but provides no evidence. What should a researcher do? (1 point)

Answer / scoring: Seek supporting evidence/corroboration and check the source before relying on it.

4. Why does an author's expertise matter? (1 point)

Answer / scoring: It helps judge whether the author is qualified on that topic; expertise alone is not proof.

5. Why use more than one source for a research explanation? (1 point)

Answer / scoring: To compare/corroborate information and obtain different relevant perspectives.

6. What should a source list help a reader do? (1 point)

Answer / scoring: Find, credit, and check the sources used.

7. What is plagiarism? (1 point)

Answer / scoring: Presenting another person's words or ideas as one's own without appropriate credit.

8. Why cite a paraphrase? (1 point)

Answer / scoring: The idea still comes from another source even though the wording changes.

9. Why is Source A a better basis for this investigation than Source B? (1 point)

Answer / scoring: A supplies dated measurements and a traceable method; B makes an unsupported absolute claim.

10. Why should researchers still check Source A's method? (1 point)

Answer / scoring: An author and date do not ensure that measurements or procedures are accurate.

11. Correct this misspelling: enviroment. (1 point)

Answer / scoring: environment

12. Write two organized paragraphs (at least 8 sentences total) explaining an important idea from the supplied passage. Paraphrase two relevant details and credit the supplied passage as your source. Include one reflection about how your interpretation might differ from another reader's. (5 points)

Answer / scoring: Rubric: clear relevant main idea/claim (1); two accurate supporting details (1 each); reasoning/organization (1); readable complete sentences and conventions (1). Accept defensible interpretations, not just the sample central idea: Credible research connects traceable evidence with cautious conclusions.

Science - 28 points

Score: _____ / 28

W1 data: A fictional class compares water collection in two labeled trials using the same time interval. Trial A: 11 milliliters of water collected; Trial B: 19 milliliters of water collected. These are invented assessment data, not a claim about a real investigation.

W2 data: A fictional class compares water collection in two labeled trials using the same time interval. Trial A: 12 milliliters of water collected; Trial B: 20 milliliters of water collected. These are invented assessment data, not a claim about a real investigation.

W3 data: A fictional class compares water collection in two labeled trials using the same time interval. Trial A: 13 milliliters of water collected; Trial B: 21 milliliters of water collected. These are invented assessment data, not a claim about a real investigation.

W4 data: A fictional class compares water collection in two labeled trials using the same time interval. Trial A: 14 milliliters of water collected; Trial B: 22 milliliters of water collected. These are invented assessment data, not a claim about a real investigation.

1. How can weather observations vary within one region? (1 point)

Answer / scoring: Time, elevation, land cover, distance to water, and other factors differ.

2. Use W1 data. Which trial has the higher recorded value, and by how much? (2 points)

Answer / scoring: Trial B; 8 milliliters of water collected.

3. Compare short-term observations with long-term climate data. (3 points)

Answer / scoring: Rubric (3): scientifically accurate parts/ideas (1); correct relationships/reasoning (1); clear labels/evidence or method (1). Correct time scales, variables, and cautious inference.

4. What is a weather front? (1 point)

Answer / scoring: A boundary between air masses with different characteristics.

5. Use W2 data. Which trial has the higher recorded value, and by how much? (2 points)

Answer / scoring: Trial B; 8 milliliters of water collected.

6. Draw two air masses and a front with possible weather changes. (3 points)

Answer / scoring: Rubric (3): scientifically accurate parts/ideas (1); correct relationships/reasoning (1); clear labels/evidence or method (1). Correct boundary and plausible temperature/moisture changes; forecast is qualified, not certain.

7. Why combine satellite, radar, and surface data? (1 point)

Answer / scoring: They provide different scales and variables, improving interpretation.

8. Use W3 data. Which trial has the higher recorded value, and by how much? (2 points)

Answer / scoring: Trial B; 8 milliliters of water collected.

9. Plan a weather-data comparison. (3 points)

Answer / scoring: Rubric (3): scientifically accurate parts/ideas (1); correct relationships/reasoning (1); clear labels/evidence or method (1). Identify each source/variable, align times/locations, and discuss limits.

10. Why evaluate forecasts after the predicted period? (1 point)

Answer / scoring: It reveals accuracy, errors, and possible model improvements.

11. Use W4 data. Which trial has the higher recorded value, and by how much? (2 points)

Answer / scoring: Trial B; 8 milliliters of water collected.

12. Create a forecast evaluation plan. (3 points)

Answer / scoring: Rubric (3): scientifically accurate parts/ideas (1); correct relationships/reasoning (1); clear labels/evidence or method (1). Prediction, verification data, comparison metric, and revision/reflection.

13. Choose one of the supplied data comparisons. Write a claim supported by those data, give the two values as evidence, and explain what further information would be needed for a causal explanation. (4 points)

Answer / scoring: Rubric (4): accurate limited claim (1); two correct data values (1); logical connection to claim (1); relevant limitation/next test (1). Accept varied claims bounded by the supplied data.

Social Studies - 24 points

Score: _____ / 24

W1 background: It continued the eastern Roman political tradition, with Constantinople as a central city. Its position connected important land/sea routes and offered strategic advantages. This is an assessment background note written for this packet, not an actual historical primary-source quotation.

W2 background: The Arabian Peninsula in the seventh century CE. Movement of people, texts, institutions, and contacts can carry ideas; mechanisms vary. This is an assessment background note written for this packet, not an actual historical primary-source quotation.

W3 background: Mathematics, medicine, astronomy, philosophy, literature, or related fields. It helped preserve, compare, adapt, and extend knowledge from different traditions. This is an assessment background note written for this packet, not an actual historical primary-source quotation.

W4 background: Travelers share ideas, techniques, beliefs, languages, and information. Power, conflict, and access can differ among participants. This is an assessment background note written for this packet, not an actual historical primary-source quotation.

1. What was the Byzantine Empire's connection to Rome? (1 point)

Answer / scoring: It continued the eastern Roman political tradition, with Constantinople as a central city.

2. Why did location matter for Constantinople? (1 point)

Answer / scoring: Its position connected important land/sea routes and offered strategic advantages.

3. Explain continuity and change in Byzantine history. (3 points)

Answer / scoring: Rubric (3): accurate relevant content (1); supported comparison/relationship (1); clear organization and source/evidence awareness (1). One Roman continuity, one change, context, and evidence.

4. Where did Islam begin as a historical religious movement? (1 point)

Answer / scoring: The Arabian Peninsula in the seventh century CE.

5. How can trade and other networks help beliefs spread? (1 point)

Answer / scoring: Movement of people, texts, institutions, and contacts can carry ideas; mechanisms vary.

6. Map a route of cultural/religious exchange. (3 points)

Answer / scoring: Rubric (3): accurate relevant content (1); supported comparison/relationship (1); clear organization and source/evidence awareness (1). Accurate broad regions, a plausible mechanism, and limits; distinguish exchange from universal acceptance.

7. Name one field developed through scholarship in the Islamic Golden Age. (1 point)

Answer / scoring: Mathematics, medicine, astronomy, philosophy, literature, or related fields.

8. Why did translation matter to scholarship? (1 point)

Answer / scoring: It helped preserve, compare, adapt, and extend knowledge from different traditions.

9. Model a scholarly exchange. (3 points)

Answer / scoring: Rubric (3): accurate relevant content (1); supported comparison/relationship (1); clear organization and source/evidence awareness (1). Text/idea, translation/contact, new use/development, and acknowledgment of multiple contributors.

10. Why can trade carry more than goods? (1 point)

Answer / scoring: Travelers share ideas, techniques, beliefs, languages, and information.

11. Why is cultural exchange not always peaceful or equal? (1 point)

Answer / scoring: Power, conflict, and access can differ among participants.

12. Explain an exchange through a historical network. (3 points)

Answer / scoring: Rubric (3): accurate relevant content (1); supported comparison/relationship (1); clear organization and source/evidence awareness (1). Goods and ideas, participants, mechanism, and a benefit/unequal effect.

13. Choose one background note. Describe an additional perspective or source that could improve it, and explain what it might reveal or challenge. (4 points)

Answer / scoring: Rubric (4): identifies chosen note/claim (1); relevant additional source/perspective (1); explains new insight/check (1); recognizes limits rather than treating one source as complete truth (1).

Spanish - 12 points

Score: _____ / 12

Spanish model and vocabulary (cue support allowed)

Estudio por la mañana. Leo por la noche.
Word bank: estudiar, leer, comer, dormir, la mañana, la noche.

1. Week 1: Translate to study into Spanish. (1 point)

Answer / scoring: estudiar

2. Week 2: Complete: Yo ___ español. (hablar, present tense) (1 point)

Answer / scoring: hablo

3. Week 3: Translate to study into Spanish. (1 point)

Answer / scoring: estudiar

4. Week 4: Translate to study into Spanish. (1 point)

Answer / scoring: estudiar

5. Read the supplied Spanish model. Give its meaning in English. (2 points)

Answer / scoring: I study in the morning. I read at night.

6. Write one new correct sentence using vocabulary from this module. (2 points)

Answer / scoring: One point for relevant vocabulary; one for a meaningful correct sentence. Accept varied answers.

7. Oral task (teacher listens): Use 6-8 short statements or a 6-line exchange about this week/module topic. You may use the supplied vocabulary/model as a cue, but add your own relevant detail or question. (4 points)

Answer / scoring: Oral rubric (4): communicates topic meaning (1); uses relevant vocabulary (1); uses understandable phrases/sentences (1); responds to teacher or adds a relevant original detail/question (1). Sample cue: Estudio por la mañana. Leo por la noche. Accept different accurate answers and appropriate gender/person forms. Minor accent/pronunciation errors do not erase understood meaning.